MATH 161, Autumn 2025
SCRIPT 2: The Rationals

Note that all of the following proofs should be straightforward consequences of properties of the integers; you should be sure to make it clear what facts about the integers you are using. See Script 0 for a list of the defining properties of $\mathbb{Z}$.

Definition 2.1. Let X be a nonempty set. A *relation* R on X is a subset of $X \times X$. The statement $(x, y) \in R$ is read as 'x is related to y by the relation R,' and is often denoted $x \sim y$.

A relation is *reflexive* if $x \sim x$ for all $x \in X$.
A relation is *symmetric* if $y \sim x$ whenever $x \sim y$.
A relation is *transitive* if $x \sim z$ whenever $x \sim y$ and $y \sim z$.
A relation is an *equivalence relation* if it is reflexive, symmetric and transitive.

Exercise 2.2. Determine which of the following are equivalence relations.

a) Any set X with the relation $=$. So $x \sim y$ if and only if $x = y$.

   Answer. True. We may say $x = x$, so it is reflective. We may say $x = y$ implies $y = x$, so it is symmetric. Finally, if $x = y$ and $y = z$, then we may say $x = z$ and therefore it is transitive. □

b) $\mathbb{Z}$ with the relation $<$.

   Answer. False. $2 \not< 2$ and hence it is not in equivalence. □

c) Any subset X of $\mathbb{Z}$ with the relation $\leq$. So $x \sim y$ if and only if $x \leq y$.

   Answer. False. It may not be symmetric. For example, $2 \leq 3$ but $3 \not\leq 2$. Therefore, it is not in equivalence. □

d) $X = \mathbb{Z}$ with $x \sim y$ if and only if $y - x$ is divisible by 5.

   Answer. This is true.

   It is reflective as $x - x = 0$ which is divisble by 5.

   It is symmetric as $y - x = -(x - y)$. Hence, if $y - x$ is divisible by 5, then $x - y$ will be divisible by 5.

   It is transitive. Say we have $y - x$, which is divisible by 5 and $z - y$, which is divisible by 5. Then we may write the two equations as $(y - x) + (z - y) = 5k + 5m$ where k and m are arbitrary numbers. Therefore, we can write as $z - x = 5(k + m)$, which is divisible by 5.

   □

e) $X = \{(a,b) \mid a,b \in \mathbb{Z}, b \neq 0\}$ with the relation $\sim$ defined by

$$(a,b) \sim (c,d) \quad \text{if and only if} \quad ad = bc.$$

Answer. It is true. We can say that $ad = ba$, so it is reflective. We may say that $ad = bc$, so therefore it is implied that $cb = da$, so it is symmetric. Lastly, if $ad = bc$ and $cf = de$ then we may say $af = be$ as $adcf = bcde$. Hence, it is transitive. Therefore, we have proven it is in equivalence. □

Remark 2.3. A *partition* of a set is a collection of non-empty disjoint subsets whose union is the original set. Any equivalence relation on a set creates a partition of that set by collecting into subsets all of the elements that are equivalent (related) to each other. When the partition of a set arises from an equivalence relation in this manner, the subsets are referred to as *equivalence classes*. (See Exercises 2 and 3 in the Additional Exercises section below.)

Remark 2.4. If we think of the set X in 2.2e) as representing the collection of all fractions whose denominators are not zero, then the relation $\sim$ may be thought of as representing the equivalence of two fractions.

Definition 2.5. As a set, the *rational numbers*, denoted $\mathbb{Q}$, are the equivalence classes in the set $X = \{(a,b) \mid a,b \in \mathbb{Z}, b \neq 0\}$ under the equivalence relation $\sim$ as defined in 2.2e). If $(a,b) \in X$, we denote the equivalence class of this element as $\left[\frac{a}{b}\right]$. So

$$\left[\frac{a}{b}\right] = \{(x_1,x_2) \in X \mid (x_1,x_2) \sim (a,b)\} = \{(x_1,x_2) \in X \mid x_1 b = x_2 a\}.$$

Then,

$$\mathbb{Q} = \left\{ \left[\frac{a}{b}\right] \mid (a,b) \in X \right\}.$$

Exercise 2.6. $\left[\dfrac{a}{b}\right] = \left[\dfrac{a'}{b'}\right] \iff (a,b) \sim (a',b')$.

Proof of 2.6. We want to prove that $\left[\frac{a}{b}\right] = \left[\frac{a'}{b'}\right]$ is equivalent to the statement $(a,b) \sim (a',b')$.
Let $x,y \in \left[\frac{a}{b}\right]$ and $\left[\frac{a}{b}\right] = \left[\frac{a'}{b'}\right]$. We prove this through double inclusion.

By **Definition 2.5**, we know that if $(a.b) \in X$, then we may state $\left[\frac{a}{b}\right]$. As $x,y \in \left[\frac{a}{b}\right]$, we may state that $(x.y) \sim (a,b)$. As $\left[\frac{a}{b}\right] = \left[\frac{a'}{b'}\right]$, we may state that $(x,y) \sim (a,b)$ and $(x,y) \sim (a',b')$, and as they have an equivalence relation, we may therefore say that $(a,b) \sim (a',b')$ through transitivity.

Therefore, we may say that $(x,y) \in \left[\frac{a'}{b'}\right]$. Hence, as $(x,y) \in \left[\frac{a'}{b'}\right]$ and $(x,y) \in \left[\frac{a}{b}\right]$, we may say that $\left[\frac{a}{b}\right] \subseteq \left[\frac{a'}{b'}\right]$.

If we repeated the process where $(x,y) \in \left[\frac{a'}{b'}\right]$, we may analogously say $\left[\frac{a'}{b'}\right] \subseteq \left[\frac{a}{b}\right]$. Hence, through double inclusive we may say that $\left[\frac{a}{b}\right] = \left[\frac{a'}{b'}\right]$.

□

Definition 2.7. We define the binary operations addition and multiplication on $\mathbb{Q}$ as follows. If $\left[\dfrac{a}{b}\right]$ and $\left[\dfrac{c}{d}\right]$ are in $\mathbb{Q}$, then:

$$\left[\frac{a}{b}\right] +_{\mathbb{Q}} \left[\frac{c}{d}\right] = \left[\frac{ad+bc}{bd}\right]$$
$$\left[\frac{a}{b}\right] \cdot_{\mathbb{Q}} \left[\frac{c}{d}\right] = \left[\frac{ac}{bd}\right].$$

We use the notation $+_{\mathbb{Q}}$ and $\cdot_{\mathbb{Q}}$ to represent addition and multiplication in $\mathbb{Q}$ so as to distinguish these operations from the usual addition $(+)$ and multiplication $(\cdot)$ in $\mathbb{Z}$.

We want to know that addition and multiplication are "well-defined", by which we mean that if we change the representatives of the classes $\left[\frac{a}{b}\right]$ and $\left[\frac{c}{d}\right]$, then does this not change the resulting classes on the right-hand-side of the equalities in the definition. To prove the next theorem first you will need to formulate a precise statement about what needs to be checked.

Theorem 2.8. *Addition and multiplication in $\mathbb{Q}$ are well-defined.*

Proof of Theorem 2.8. We want to prove that the addition and multiplication of $\mathbb{Q}$ is well-defined.

Let $(a, b), (c, d) \in X$ and $(a, b) \sim (a', b')$ and $(c, d) \sim (c'd')$
Suppose that we have the expression $adb'd' + bcb'd'$ We must prove that $adb'd' + bcb'd' = a'd'bd + b'c'bd'$ to thereby prove that addition in $\mathbb{Q}$ is well-defined.

Through the equivalences of $(a, b) \sim (a', b')$ and $(c, d) \sim (c'd')$, we may write $adb'd' + bcb'd$ as $(a'b)dd' + (b'c)dd'$. This equals the expression $a'd'bd + b'c'bd'$. Therefore $adb'd' + bcb'd' = a'd'bd + b'c'bd'$.

We way rewrite the equation as $(ab + bc)(b'd') = (a'd' + b'c')(bd)$, by factorising. This leads us to $\left[\frac{ad+bc}{bd}\right] = \left[\frac{a'd'+b'c'}{b'd'}\right]$. This, thereby, proves that additon is well defined.

Similarly we must prove that multiplication is well-defined. Let there be an expression $(ac)(b'd')$. Through the equivalences, we may rewrite $(ac)(b'd')$ as $(a'c')(bd)$. As they are equivalent, we may write them in an equation as follows

$$(ac)(b'd') = (a'c')(bd)$$

By factoring on either side we get the expression

$$\left[\frac{ac}{bd}\right] = \left[\frac{a'c'}{b'd'}\right]$$

As these are equivalent, we may say that multiplication is well defined.

$\square$

Theorem 2.9. *a)* **Commutativity of addition**

$\left[\dfrac{a}{b}\right] +_{\mathbb{Q}} \left[\dfrac{c}{d}\right] = \left[\dfrac{c}{d}\right] +_{\mathbb{Q}} \left[\dfrac{a}{b}\right]$ *for all* $\left[\dfrac{a}{b}\right], \left[\dfrac{c}{d}\right] \in \mathbb{Q}$.

Proof of a. Suppose we take each of the two expressions separately.

$$\left[\frac{a}{b}\right] +_{\mathbb{Q}} \left[\frac{c}{d}\right] = \left[\frac{ad+bc}{bd}\right]$$
$$\left[\frac{c}{d}\right] +_{\mathbb{Q}} \left[\frac{a}{b}\right] = \left[\frac{da+cb}{db}\right]$$

As the sets $(a,b),(c,d) \in \mathbb{Q}$, we may perform integer operators on the variable. Hence, the order of multiplication and addition does not matter. Therefore, we may say that $\left[\frac{ad+bc}{bd}\right] = \left[\frac{da+cd}{db}\right]$ proves the commutativity of addition. □

b) **Associativity of addditon**

$$\left(\left[\frac{a}{b}\right] +_{\mathbb{Q}} \left[\frac{c}{d}\right]\right) +_{\mathbb{Q}} \left[\frac{e}{f}\right] = \left[\frac{a}{b}\right] +_{\mathbb{Q}} \left(\left[\frac{c}{d}\right] +_{\mathbb{Q}} \left[\frac{e}{f}\right]\right) \textit{ for all } \left[\frac{a}{b}\right], \left[\frac{c}{d}\right], \left[\frac{e}{f}\right] \in \mathbb{Q}.$$

Proof of b. We can rewrite left hand side through $(\left[\frac{a}{b}\right] +_{\mathbb{Q}} \left[\frac{c}{d}\right]) +_{\mathbb{Q}} \left[\frac{e}{f}\right] = \left[\frac{ad+cb}{bd}\right] +_{\mathbb{Q}} \left[\frac{e}{f}\right] = \left[\frac{adf+bcf+bdc}{bdf}\right]$

We can rewrite the right hand side through $\left[\frac{a}{b}\right] +_{\mathbb{Q}} (\left[\frac{c}{d}\right] +_{\mathbb{Q}} \left[\frac{e}{f}\right]) = \left[\frac{a}{b}\right] +_{\mathbb{Q}} \left[\frac{ef+de}{df}\right] = \left[\frac{bcf+bde+adf}{bdf}\right]$

As the ordered pairs $(a,b),(c,d),(e,f) \in \mathbb{Q}$, we may perform integer operators on the variable. Hence, the order of multiplication and addition does not matter. Therefore, we may say that $\left[\frac{ad+bc}{bd}\right] = \left[\frac{da+cd}{db}\right]$ proves the commutativity of addition.

□

c) **Existence of an additive identity**

$$\left[\frac{a}{b}\right] +_{\mathbb{Q}} \left[\frac{0}{1}\right] = \left[\frac{a}{b}\right], \textit{ for all } \left[\frac{a}{b}\right] \in \mathbb{Q}.$$

Proof of c. We define $\left[\frac{0}{1}\right]$ to be $(0,1)$ and contains no value. By **Definition 2.7** we can perform binary additon as follows

$$\left[\frac{a}{b}\right] +_{\mathbb{Q}} \left[\frac{0}{1}\right] = \left[\frac{(a)(1)+(b)(0)}{(b)(1)}\right] = \left[\frac{a}{b}\right]$$

Therefore, this proves the existence of an additive identity. □

d) **Existence of additive inverses** $\left[\frac{a}{b}\right] +_{\mathbb{Q}} \left[\frac{-a}{b}\right] = \left[\frac{0}{1}\right]$, *for all* $\left[\frac{a}{b}\right] \in \mathbb{Q}$.

Proof of d.
$$\left[\frac{a}{b}\right] +_{\mathbb{Q}} \left[\frac{-a}{b}\right] = \left[\frac{(a)(b) + (-a)(b)}{b^2}\right] = \left[\frac{ab - ab}{b^2}\right] = \left[\frac{0}{b^2}\right]$$

As $\left[\frac{0}{b^2}\right]$ is $(0, b^2)$ and equals 0. Therefore, we may say it is in equivalence with $(0, 1)$ and therefore $\left[\frac{a}{b}\right] +_{\mathbb{Q}} \left[\frac{-a}{b}\right] = \left[\frac{0}{1}\right]$. □

e) **Commutativity of multiplication**
$\left[\frac{a}{b}\right] \cdot_{\mathbb{Q}} \left[\frac{c}{d}\right] = \left[\frac{c}{d}\right] \cdot_{\mathbb{Q}} \left[\frac{a}{b}\right]$ *for all* $\left[\frac{a}{b}\right], \left[\frac{c}{d}\right] \in \mathbb{Q}$.

Proof of e. Firstly, we may expand the left-hand side as follows
$$\left[\frac{a}{b}\right] \cdot_{\mathbb{Q}} \left[\frac{c}{d}\right] = \left[\frac{ac}{bd}\right]$$

Secondly, we may expand the right-hand side as follows
$$\left[\frac{c}{d}\right] \cdot_{\mathbb{Q}} \left[\frac{a}{b}\right] = \left[\frac{ca}{db}\right]$$

By **Definition 2.7**, as we are using the operation of multiplication, the order of the variables does not affect the product. Therefore, we may say that $\left[\frac{a}{b}\right] \cdot_{\mathbb{Q}} \left[\frac{c}{d}\right] = \left[\frac{c}{d}\right] \cdot_{\mathbb{Q}} \left[\frac{a}{b}\right]$.

□

f) **Associativity of multiplication**
$\left(\left[\frac{a}{b}\right] \cdot_{\mathbb{Q}} \left[\frac{c}{d}\right]\right) \cdot_{\mathbb{Q}} \left[\frac{e}{f}\right] = \left[\frac{a}{b}\right] \cdot_{\mathbb{Q}} \left(\left[\frac{c}{d}\right] \cdot_{\mathbb{Q}} \left[\frac{e}{f}\right]\right)$ *for all* $\left[\frac{a}{b}\right], \left[\frac{c}{d}\right], \left[\frac{e}{f}\right] \in \mathbb{Q}$.

Proof of f. Firstly, we may rewrite the left-hand side as
$$(\left[\frac{a}{b}\right] \cdot_{\mathbb{Q}} \left[\frac{c}{d}\right]) \cdot_{\mathbb{Q}} \left[\frac{e}{f}\right] = \left[\frac{(a)(c)}{(b)(d)}\right] \cdot_{\mathbb{Q}} \left[\frac{e}{f}\right] = \left[\frac{ace}{bdf}\right]$$

Secondly, we may rewrite the right-hand side as
$$\left[\frac{a}{b}\right] \cdot_{\mathbb{Q}} (\left[\frac{c}{d}\right] \cdot_{\mathbb{Q}} \left[\frac{e}{f}\right]) = \left[\frac{a}{b}\right] \cdot_{\mathbb{Q}} \left[\frac{(c)(e)}{(d)(f)}\right] = \left[\frac{ace}{bdf}\right]$$

As the left-hand side expansion is in equivalence with the right-hand side expansion, we have proven the associativity of multiplication.

□

g) **Existence of a multiplicative identity**
$\left[\frac{a}{b}\right] \cdot_{\mathbb{Q}} \left[\frac{1}{1}\right] = \left[\frac{a}{b}\right]$, *for all* $\left[\frac{a}{b}\right] \in \mathbb{Q}$.

Proof of g. We want to prove the existence of a multiplicative identity through direct proof. We may define $\left[\frac{1}{1}\right]$ to be $(1,1)$ and function as an element. Therefore, by **Definition 2.7**, we may write

$$\left[\frac{a}{b}\right] \cdot_{\mathbb{Q}} \left[\frac{1}{1}\right] = \left[\frac{(a)(1)}{(b)(1)}\right] = \left[\frac{a}{b}\right]$$

Therefore, we have proved the existence of a multiplicative identity. □

h) **Existence of multiplicative inverses for nonzero elements**
$\left[\frac{a}{b}\right] \cdot_{\mathbb{Q}} \left[\frac{b}{a}\right] = \left[\frac{1}{1}\right]$, *for all* $\left[\frac{a}{b}\right] \in \mathbb{Q}$ *such that* $\left[\frac{a}{b}\right] \neq \left[\frac{0}{1}\right]$.

Proof of h. .

By **Definition 2.7** we may use the binary operations for mulitplication and therefore we may write

$$\left[\frac{a}{b}\right] \cdot_{\mathbb{Q}} \left[\frac{b}{a}\right] = \left[\frac{(a)(b)}{(b)(a)}\right] = \left[\frac{ab}{ba}\right]$$

AS $\left[\frac{ab}{ba}\right] = \left[\frac{1}{1}\right]$, we have proven the existence of multiplicative inverse for nonzero elements. □

i) **Distributivity**
$\left[\frac{a}{b}\right] \cdot_{\mathbb{Q}} \left(\left[\frac{c}{d}\right] +_{\mathbb{Q}} \left[\frac{e}{f}\right]\right) = \left(\left[\frac{a}{b}\right] \cdot_{\mathbb{Q}} \left[\frac{c}{d}\right]\right) +_{\mathbb{Q}} \left(\left[\frac{a}{b}\right] \cdot_{\mathbb{Q}} \left[\frac{e}{f}\right]\right)$, *for all* $\left[\frac{a}{b}\right], \left[\frac{c}{d}\right], \left[\frac{e}{f}\right] \in \mathbb{Q}$.

Proof. We want to prove this by direct proof. If we start with the expression on the left-hand side, we get

$$\left[\frac{a}{b}\right] \cdot_{\mathbb{Q}} \left(\left[\frac{c}{d}\right] +_{\mathbb{Q}} \left[\frac{e}{f}\right]\right) = \left[\frac{cf+de}{df}\right] \cdot_{\mathbb{Q}} \left[\frac{a}{b}\right]$$

$$\left[\frac{cf+de}{df}\right] \cdot_{\mathbb{Q}} \left[\frac{a}{b}\right] = \left[\frac{acf+ade}{bdf}\right] = \left[\frac{acbf+aebd}{bdbf}\right]$$

$$\left[\frac{acbf+aebd}{bdbf}\right] = \left[\frac{ac}{bd}\right] + \left[\frac{ae}{bf}\right] = \left(\left[\frac{a}{b}\right] \cdot_{\mathbb{Q}} \left[\frac{c}{d}\right]\right) +_{\mathbb{Q}} \left(\left[\frac{a}{b}\right] \cdot_{\mathbb{Q}} \left[\frac{e}{f}\right]\right)$$

Therefore, we have proved distributivity.

□

Theorem 2.10. $\mathbb{Q}$ *is countable.*

Hint: look back at Script 1

Proof. Let $f : \mathbb{Q} \to \mathbb{Z} \times \mathbb{N}$ be defined as $f(\left[\frac{a}{b}\right] = (a.b)$, for $\left[\frac{a}{b}\right] \in \mathbb{Q}$ where $a \in \mathbb{Z}$ and $b \in \mathbb{N}$. As we have established that $\mathbb{Z} \times \mathbb{N}$, we want to prove that f is injective.

Case 1: $a \neq a', b \neq$. Then $f(\left[\frac{a}{b}\right] = (a, b) \neq (a, b') = f(\left[\frac{a'}{b'}\right])$.

Case 2: $a \neq a'$, $b \neq b'$. Then $f(\left[\frac{a}{b}\right] = (a, b) \neq (a', b) = f(\left[\frac{a'}{b'}\right])$.

Case 3: $a \neq a', b \neq b'$. Then $f(\left[\frac{a}{b}\right] = (a, b) \neq (a', b') = f(\left[\frac{a'}{b'}\right])$.

Therefore, we may say $\mathbb{Q}$ is countable. □

We will now lose the equivalence class notation and simply refer to elements of $\mathbb{Q}$ as usual. So for example, if we refer to 0, we really mean $[\frac{0}{1}]$, but this distinction should no longer be necessary or relevant.

Additional Exercises

In all exercises you are expected to prove your answer, unless explicitly stated otherwise.

1. Prove that for every $q = \left[\frac{a}{b}\right] \in \mathbb{Q}$, there is exactly one element $(a_0, b_0) \in q$ such that a_0 and b_0 have no common factors and $b_0 > 0$.

2. Let $\sim$ be an equivalence relation on X. Let $[x]$ be the equivalence class of x. Show that

   (a) $[x] = [y]$ if and only if $x \sim y$

   (b) for all $x, y \in X$,
   $$[x] \cap [y] = \begin{cases} [x] = [y] & \text{if } x \sim y \\ \emptyset & \text{if } x \not\sim y. \end{cases}$$

   (c) $X = \bigcup_{x \in X} [x]$

3. For the equivalence relations found in Exercise 2.2, what are the equivalence classes?